Label-efficient Land Cover Classification at High Spatial Resolution using Multi-stage Pre-training

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Overview

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1. Background
2. Methodology
3. Results
4. Conclusion

Land Cover

- **Land cover**: biophysical material covering the Earth's surface (e.g., water, vegetation, soil, structures, etc.) [1] **"What is there?"**

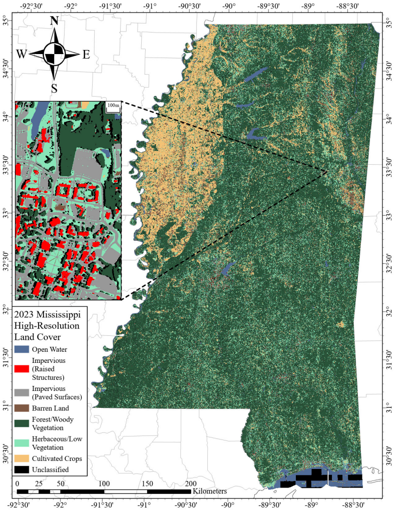
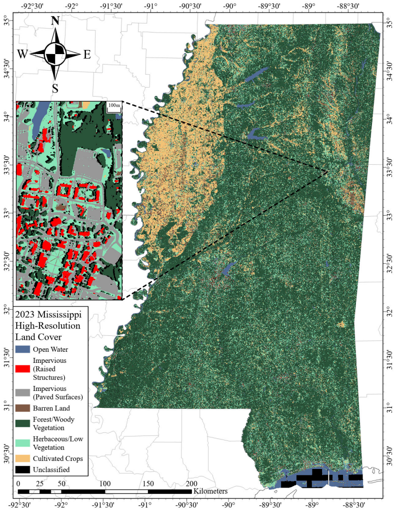


Figure 1: Land cover map of Mississippi.

Land Cover

- **Land cover**: biophysical material covering the Earth's surface (e.g., water, vegetation, soil, structures, etc.) [1] **"What is there?"**
- Land cover data is used in a wide range of disciplines, including agriculture, forestry, urban planning, and environmental science [2–4]



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- Non-parametric, supervised classifiers are often used (e.g., Decision Trees, Random Forests, Support Vector Machines, etc.) [5, 6].

Pixel-wise Classification

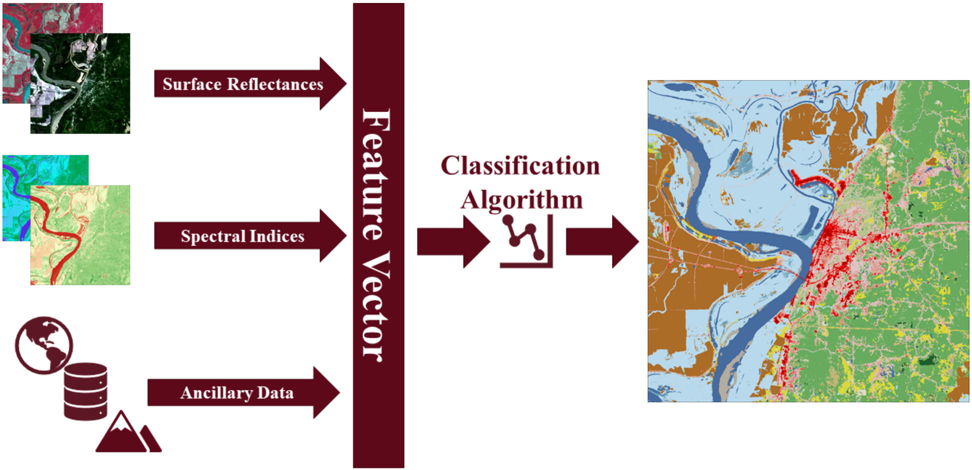

Figure 2: Pixel-wise classification workflow.

Land Cover at High Spatial Resolution

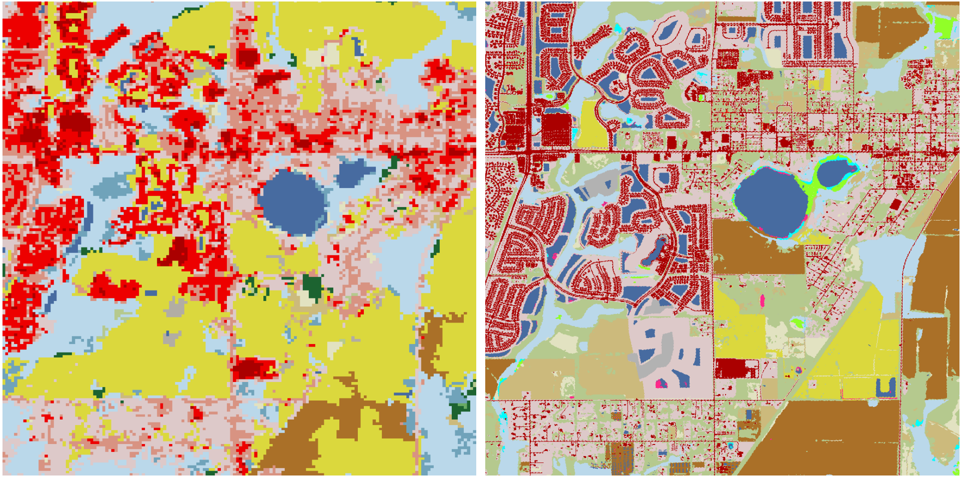

Figure 3: Comparison of land cover maps at 30 m (left) and 1 m (right) spatial resolution.

Decreased Class Discriminability

Classifying pixels gets **more difficult** as spatial resolution **increases** [7, 8].

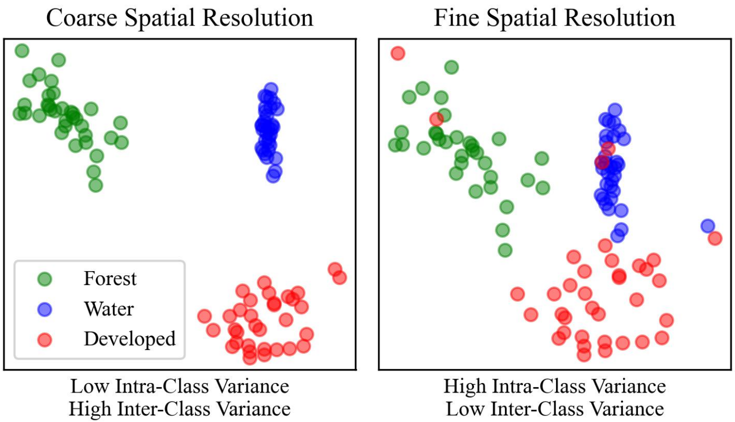

Figure 4: As spatial resolution increases, inter-class variance decreases [8].

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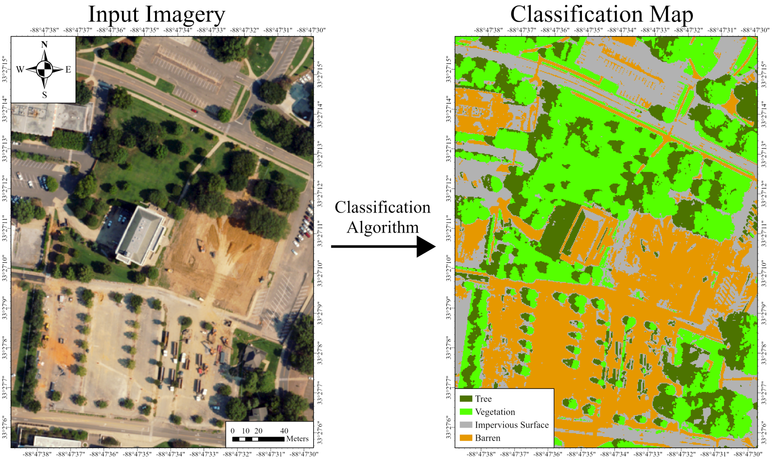

Figure 5: Applying a pixel-wise classifier to high spatial resolution data typically results in noisy classification maps.

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  3. **End-to-end learning** — unlike traditional methods, deep models can be trained in an end-to-end fashion, where little to no pre-processing or post-processing is required [11–14].

Convolutional Neural Networks (CNNs)

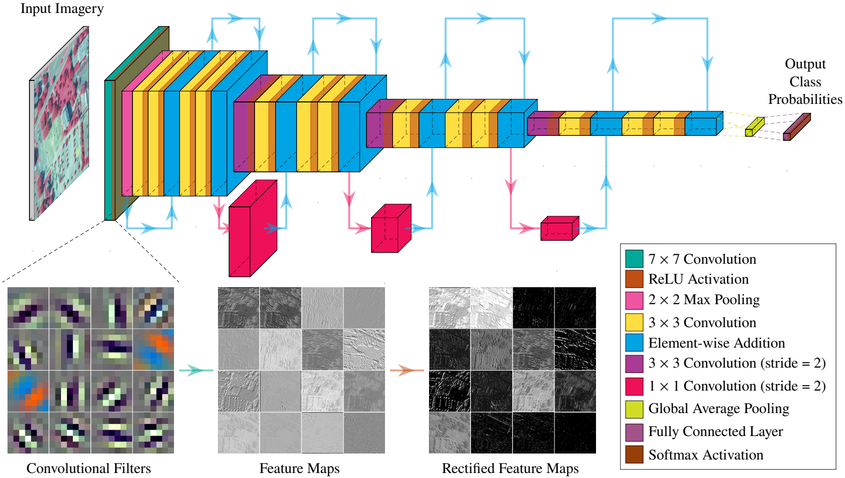

Figure 6: A ResNet-18 CNN architecture [15]. A sampling of learned convolutional filters from the first layer is shown.

U-Net

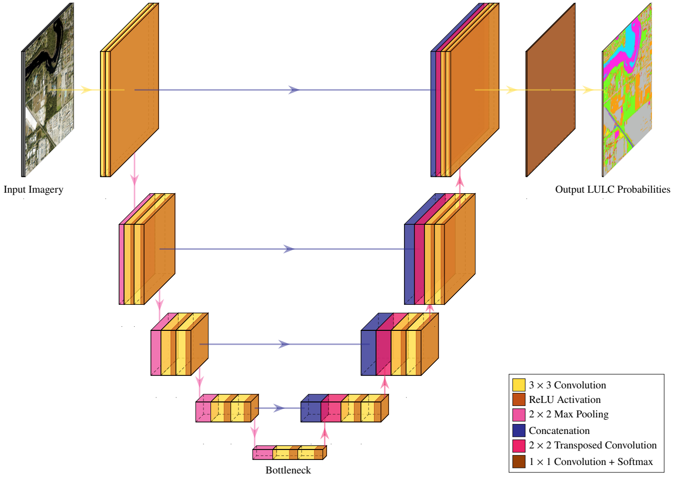

Figure 7: U-Net architecture [16].

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- For land cover, this is especially problematic, as high resolution data is complex and difficult to annotate at scale [17].
- One approach to mitigate this issue is to use **transfer learning**, where a network can be **pre-trained** on a related task with a large amount of data, then **fine-tuned** on the target task with a smaller amount of data [18].

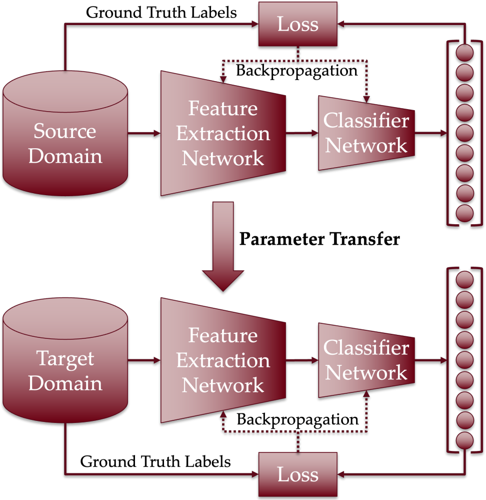

Figure 8: Transfer learning paradigm.

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- In effect, noise is added to an input image, then train the network to predict the original image from the noisy image.
- The network needs to learn **features** that are **robust** to noise while being meaningful for reconstruction [21].

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- Data annotation is expensive and time-consuming, so the main focus of our work is to devise a creative pre-training strategy that combines self-supervised pre-training with a supervised fine-tuning stage to reduce the amount of labeled data required for training a deep model.
- We construct a large U-Net model that undergoes various pre-training regimens, then fine-tune the model on only **119** tiles of fully annotated data from the state of Mississippi.

Overview

Our U-Net

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- In place of the original encoder network, we use a ResNet-152 architecture [15].
- Overall, our complete U-Net architecture has over 100,000,000 parameters, making it a powerful model, but also difficult to train — especially in the absence of large amounts of data.

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- We then place the encoder in a U-Net architecture and train the network on a moderate amount of labeled data from another region where high-resolution data is available.
- Finally, the U-Net is fine-tuned on our limited amount of labeled data from Mississippi.

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  - For fine-tuning, 250 tiles total were selected to be labeled (119 in training set, 61 in validation set, and 70 in test set)

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- 10,220 tiles used for training and 5,110 for validation.

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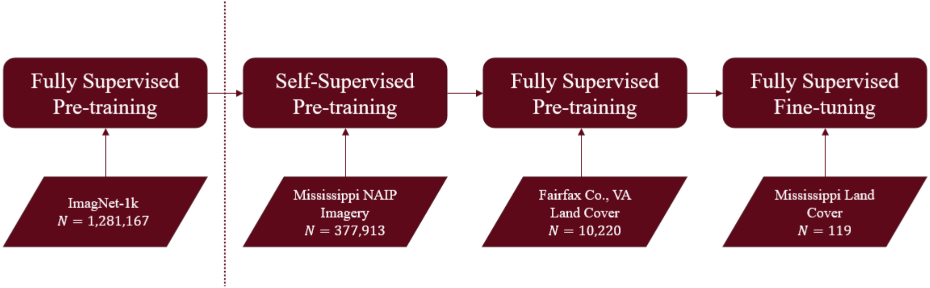

Figure 9: Model training flowchart.

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Overview

Pre-training Strategies

Pre-training with a denoising autoencoder improved the performance of the model under the 2-stage pre-training strategy.

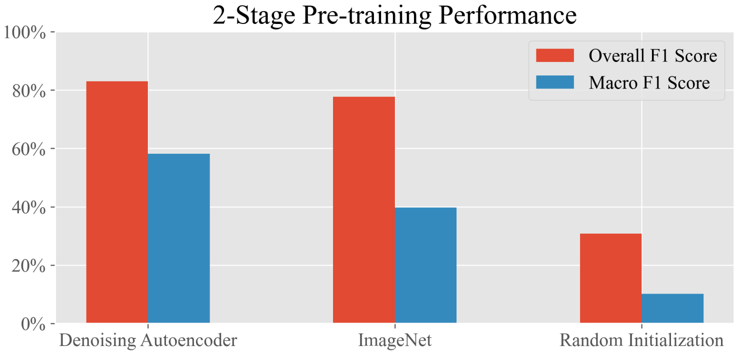

Figure 10: Performance of models on the Mississippi land cover dataset.

Pre-training Strategies

An additional fully supervised pre-training stage substantially improves the performance of the model.

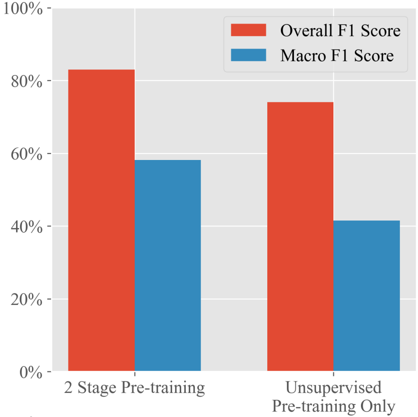

Figure 11: Performance of the best model with and without additional supervised pre-training.

Confusion Matrix

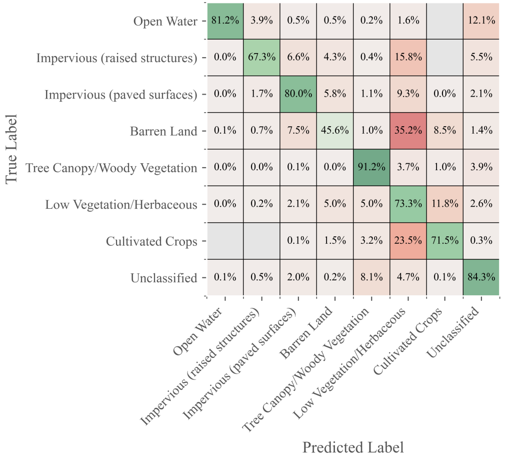

Figure 12: Confusion matrix of the final model on the Mississippi land cover dataset.

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Mississippi Hi-Res Land Cover

Figure 13: 2023 Mississippi Land Cover Map.

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Summary

- Large scale pre-training of a deep model using an unsupervised regimen can enable the training of an over-parameterized deep model on a small amount of labeled data.
- Further, additional pre-training using a modest amount of labeled data can boost the performance of the model even further.
- Still, classifying more visually intricate land cover classes remains a challenge, even with a large model.

Future Works

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- Applying model to super-resolved satellite imagery.

Acknowledgements

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